

Question Number	Answer	Additional guidance	Mark
4(a)(i)	Archaea and Eukarya	mark first 2 answers only both required for the mark irrespective of which line accept eukaryote / eukaryotes/ eukaryota etc accept phonetic spellings	(1)

Question Number	Answer	Additional guidance	Mark
4(a)(ii)	A calculation in which: • correct calculation (1) • conversion to standard form (1)	<u>Example of calculation:</u> 20 000 000 ÷ 300 = 66666.66667 or 20 ÷ (3 × 10⁻⁴) = 66666.66667 mark for either LHS or RHS of calculation watch out as some convert both numbers to a common unit for some reason! = 6.7 × 10⁴ / 6.67 × 10⁴ / 7 × 10⁴ 6.667 × 10⁴ / 6.6667 × 10⁴ are 1 mark as too many dp factor of 10 error in standard form = 1 mark (must be 6.7 or 7 × 10^x NOT 0.67/eq) 1 mark for 1.5 × 10⁻⁵ no credit for any other incorrect answer converted into standard form Correct answer scores full marks	(2)

most common responses – but please follow MS at all times for other answers.

0	1	2
6.7	20 000 000 ÷ 300	6.7 × 10 ⁴
7	20 ÷ (3 × 10 ⁻⁴)	6.67 × 10 ⁴
6.6 × 10 ^{incorrect number}	66666.66667 or correctly rounded to any or no dp e.g. 66667 etc	7 × 10 ⁴ OR 7.0 × 10 ⁴
6667	1.5 × 10 ⁻⁵	
	{7 or 6.7 or 6.67 or other correct} × 10 ^x	
	6.6 × 10 ⁴	
	6.667 × 10 ⁴	
	6.6667 × 10 ⁴	

Question Number	Answer	Additional guidance	Mark
4(b)	An answer that makes reference to the following points: • (because circular DNA) contains {genes / alleles/ bases/ genetic information} which will be {copied onto mRNA / transcribed / codes for a protein / codes for a sequence of amino acids} (1) • (because) ribosomes will translate mRNA base sequence into an amino acid sequence (1) • correct role of membrane (1)	allow weird spellings of pepins e.g. pepsin accept where transcription occurs / (pre-) mRNA formation accept contains {genes / alleles/ bases/ genetic information} ignore codes for an amino acid accept translation occurs (at the ribosomes) / description of translation / formation of peptide bonds ignore protein synthesis as in Q stem reject transcription e.g. partially permeable / {compartmentalise / encloses} the {DNA / ribosomes} from the cytoplasm / formation of vesicles / fusing of vesicles to release proteins / control what {enters/leaves} (pepins) unqualified ignore controls what {enters/leaves} cell ignore protects {DNA / ribosomes} reject exocytosis out of cell but accept exocytosis unqualified	(3)

Question Number	Answer	Additional guidance	Mark
4(c)(i)	An explanation that makes reference to the following points: • (optimum temperature) for fastest {enzyme / metabolic} reaction (1) • (water) for {hydrolysis (reactions) / solvent / prevent dehydration} (1)	accept {faster/optimum/ increase? / more / peak} {enzyme rate of reaction / product formation / collisions between enzyme and substrate} ignore for enzyme reactions to occur unqualified accept prevent denaturing ignore transport ignore photosynthesis / respiration accept contains (dissolved) oxygen accept maintain {turgidity / rigidity}	(2)

Question Number	Answer	Additional guidance	Mark
4(c)(ii)	An explanation that makes reference to three of the following points: • as oxygen percentage increases the growth rate of A decreases whereas B increases (1) • (B growth rate increases because) it is an (obligate) aerobe} / performs aerobic respiration / it requires oxygen for (aerobic) respiration / (1) • Growth rate of type B levels off because {oxygen is in excess / another factor is limiting growth} (1) • (A growth rate decreases because) it is an anaerobe / A respire anaerobically / high oxygen percentage {inhibits growth (of A) / is toxic (to A)} / (1)	piece together accept converse for lower oxygen concentrations accept correct correlations ignore requires oxygen {unqualified / for growth} accept A is outcompeted by B ignore does not require oxygen {unqualified / for growth} ignore doesn't need oxygen for metabolic reactions ignore it lives in anaerobic conditions	(3)